

# LNP<sup>TM</sup> THERMOCOMP<sup>TM</sup> COMPOUND DX11355

REGION ASIA

## DESCRIPTION

This is a PC based compound with good plating, surface, mechanical performance and flame retardant (Halogen free), a good candidate for Laser Direct Structuring applications.

## TYPICAL PROPERTY VALUES

Revision 20190328

| PROPERTIES                                          | TYPICAL VALUES                    | UNITS             | TEST METHODS |
|-----------------------------------------------------|-----------------------------------|-------------------|--------------|
| <b>MECHANICAL</b>                                   |                                   |                   |              |
| Tensile Stress, yld, Type I, 50 mm/min              | 58                                | MPa               | ASTM D 638   |
| Tensile Stress, brk, Type I, 50 mm/min              | 52                                | MPa               | ASTM D 638   |
| Tensile Strain, yld, Type I, 50 mm/min              | 5.4                               | %                 | ASTM D 638   |
| Tensile Strain, brk, Type I, 50 mm/min              | 70                                | %                 | ASTM D 638   |
| Tensile Modulus, 50 mm/min                          | 2600                              | MPa               | ASTM D 638   |
| Flexural Stress, yld, 1.3 mm/min, 50 mm span        | 90                                | MPa               | ASTM D 790   |
| Flexural Stress, brk, 1.3 mm/min, 50 mm span        | 89                                | MPa               | ASTM D 790   |
| Flexural Modulus, 1.3 mm/min, 50 mm span            | 2400                              | MPa               | ASTM D 790   |
| <b>IMPACT</b>                                       |                                   |                   |              |
| Izod Impact, notched, 23°C                          | 800                               | J/m               | ASTM D 256   |
| <b>THERMAL</b>                                      |                                   |                   |              |
| HDT, 1.82 MPa, 3.2mm, unannealed                    | 114                               | °C                | ASTM D 648   |
| CTE, -40°C to 40°C, flow                            | 6.2E-05                           | 1/°C              | ASTM E 831   |
| CTE, -40°C to 40°C, xflow                           | 6.7E-05                           | 1/°C              | ASTM E 831   |
| Relative Temp Index, Elec <sup>(1)</sup>            | 80                                | °C                | UL 746B      |
| Relative Temp Index, Mech w/impact <sup>(1)</sup>   | 80                                | °C                | UL 746B      |
| Relative Temp Index, Mech w/o impact <sup>(1)</sup> | 80                                | °C                | UL 746B      |
| <b>PHYSICAL</b>                                     |                                   |                   |              |
| Density                                             | 1.27                              | g/cm <sup>3</sup> | ASTM D 792   |
| Moisture Absorption, 50% RH, 24 hrs                 | 0.01                              | %                 | ASTM D 570   |
| Mold Shrinkage, flow, 24 hrs                        | 0.5                               | %                 | ASTM D 955   |
| Mold Shrinkage, xflow, 24 hrs                       | 0.5                               | %                 | ASTM D 955   |
| Melt Flow Rate, 300°C/1.2 kgf                       | 12                                | g/10 min          | ASTM D 1238  |
| <b>ELECTRICAL</b>                                   |                                   |                   |              |
| Relative Permittivity, 1 GHz                        | 2.92                              | -                 | IEC 60250    |
| Dissipation Factor, 1 GHz                           | 0.007                             | -                 | IEC 60250    |
| <b>FLAME CHARACTERISTICS <sup>(1)</sup></b>         |                                   |                   |              |
| UL Yellow Card Link                                 | <a href="#">E207780-101214685</a> | -                 | -            |
| UL Recognized, 94V-0 Flame Class Rating             | ≥0.6                              | mm                | UL 94        |
| <b>INJECTION MOLDING</b>                            |                                   |                   |              |
| Drying Temperature                                  | 100 – 110                         | °C                |              |
| Drying Time                                         | 3 – 4                             | hrs               |              |
| Drying Time (Cumulative)                            | 8                                 | hrs               |              |

| PROPERTIES                  | TYPICAL VALUES | UNITS | TEST METHODS |
|-----------------------------|----------------|-------|--------------|
| Maximum Moisture Content    | 0.02           | %     |              |
| Melt Temperature            | 275 – 300      | °C    |              |
| Nozzle Temperature          | 275 – 300      | °C    |              |
| Front - Zone 3 Temperature  | 260 – 300      | °C    |              |
| Middle - Zone 2 Temperature | 255 – 295      | °C    |              |
| Rear - Zone 1 Temperature   | 250 – 290      | °C    |              |
| Hopper Temperature          | 60 – 80        | °C    |              |
| Mold Temperature            | 60 – 90        | °C    |              |
| Back Pressure               | 0.3 – 0.7      | MPa   |              |
| Screw Speed                 | 40 – 70        | rpm   |              |
| Shot to Cylinder Size       | 30 – 80        | %     |              |
| Vent Depth                  | 0.038 – 0.076  | mm    |              |

(1) UL Ratings shown on the technical datasheet might not cover the full range of thicknesses and colors. For details, please see the UL Yellow Card.

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